

# 1 Parcel Modelling

## Cloud formation and aerosol activation in a rising air parcel

### 1.1 Temperature lapse rate of the parcel

**Exercise 1.** Estimate how rapidly temperature decreases with height up to about 800 m.

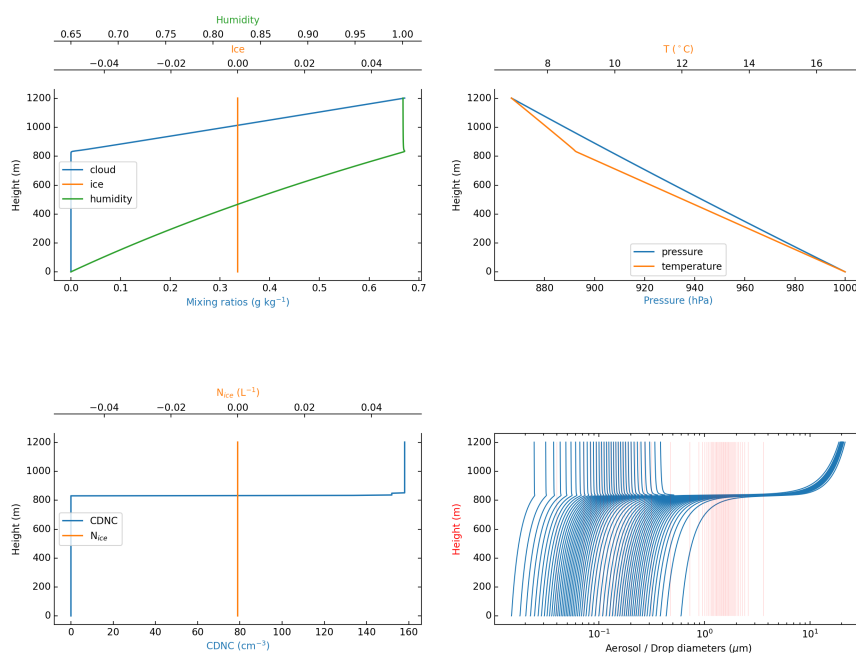
**Model answer.** From the plot, temperature falls from about 17 °C at the surface to roughly 9–9.5 °C near 800 m. This corresponds to a lapse rate of about

$$\Gamma \approx \frac{7.5-8 \text{ K}}{0.8 \text{ km}} \approx 9-10 \text{ K km}^{-1},$$

close to the dry-adiabatic value  $g/c_p \approx 9.8 \text{ K km}^{-1}$ .

**Exercise 2.** What happens to the rate of cooling above about 800 m, and why?

**Model answer.** The temperature decreases more slowly with height once cloud forms. Above cloud base, water vapour condenses and releases latent heat. This partially offsets the cooling caused by expansion, so the parcel follows a moist rather than dry adiabatic lapse rate.



**Figure 1:** Default parcel calculation. Cloud begins at about 0.8 km, where the lapse rate changes and liquid water appears.

### 1.2 Effect of vertical wind speed

**Exercise 3.** How does the lapse rate between 0 and 800 m compare between the slow, default and fast-ascent simulations?

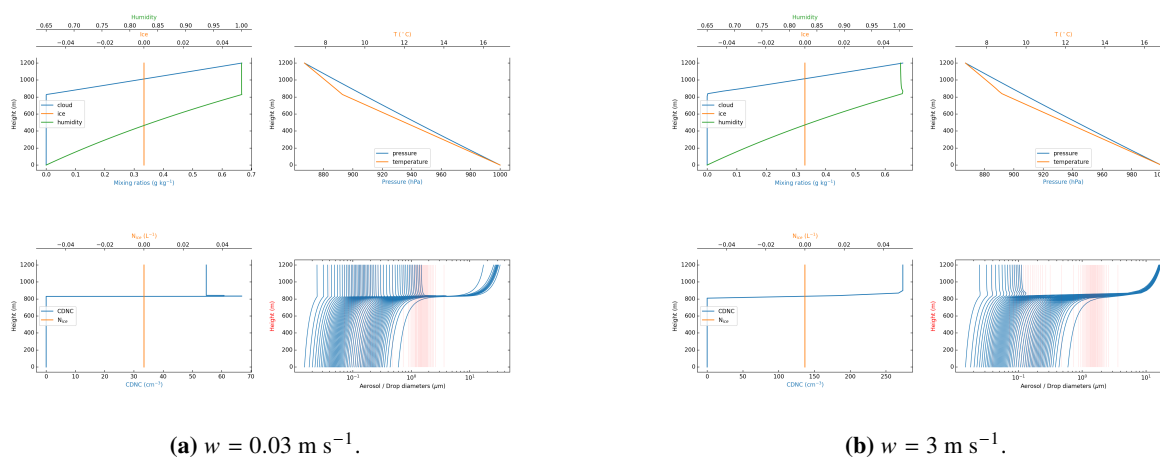
**Model answer.** It is almost unchanged. Before saturation, the parcel is approximately dry adiabatic, so temperature as a function of height is controlled mainly by thermodynamics rather than by how quickly that height is reached. Changing vertical velocity changes the time available for microphysical adjustment, but it does not substantially change the dry-adiabatic lapse rate.

**Exercise 4.** What effect does vertical velocity have on liquid-water mixing ratio and cloud-droplet number concentration (CDNC), and why?

**Model answer.** The most obvious response is in CDNC. The slow  $0.03 \text{ m s}^{-1}$  case activates only about  $60 \text{ cm}^{-3}$ , the default case about  $150\text{--}160 \text{ cm}^{-3}$ , and the  $3 \text{ m s}^{-1}$  case roughly  $270 \text{ cm}^{-3}$ . Faster ascent cools the parcel more rapidly in time, producing a larger peak supersaturation before condensation can remove the excess vapour. More, and smaller, aerosol particles can therefore activate. The total liquid-water mixing ratio is much less sensitive because it is constrained mainly by the parcel thermodynamics once saturation is reached.

**Exercise 5.** At approximately what height does cloud start to form in the three vertical-velocity experiments, and why?

**Model answer.** Cloud begins at approximately  $800\text{--}830 \text{ m}$  in all three runs. The initial humidity and thermodynamic profile are unchanged, so the parcel reaches saturation after approximately the same amount of lifting. Vertical velocity changes how fast the parcel gets there, but not greatly the height at which saturation is first reached.

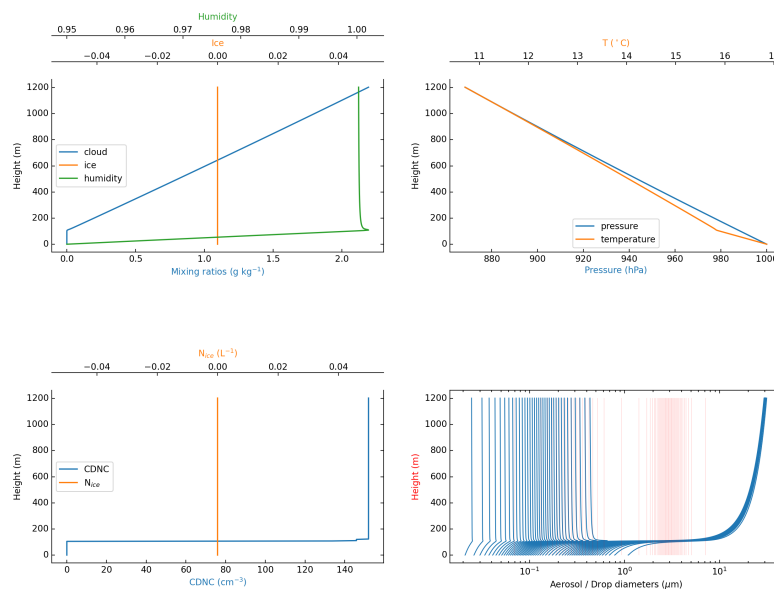


**Figure 2:** Vertical velocity has little effect on the dry lapse rate or cloud-base height, but strongly alters droplet activation.

### 1.3 Increasing humidity

**Exercise 6.** How does increasing the initial relative humidity to 0.95 change the result, and why?

**Model answer.** Cloud forms much lower, at only about  $100 \text{ m}$  rather than about  $800 \text{ m}$ . Because the parcel starts much closer to saturation, little lifting and cooling are required before condensation begins. The parcel therefore spends a much greater vertical distance in cloud and accumulates considerably more liquid water by  $1200 \text{ m}$ . Latent heating also begins much earlier, so the temperature profile departs from the dry adiabat at a much lower altitude.



**Figure 3:** With initial RH = 0.95, saturation is reached after only a small amount of lifting and the cloud is much deeper.

## 1.4 Adding sea-salt aerosol

**Exercise 7.** What is the effect of adding  $1000 \times 10^6 \text{ kg}^{-1}$  sea-salt particles with the default 100 nm median diameter and width?

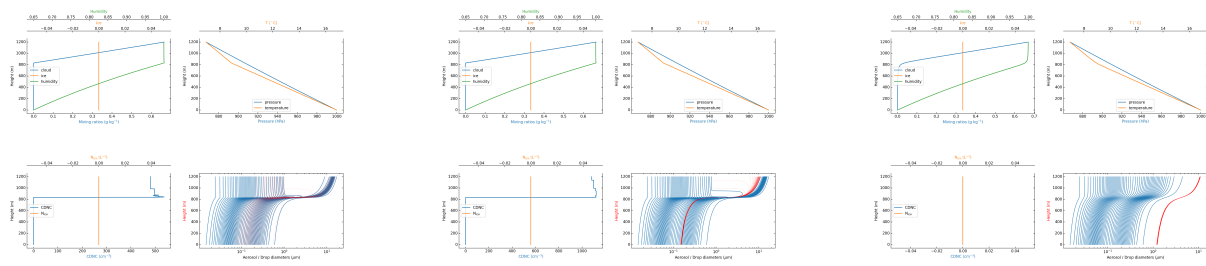
**Model answer.** The CDNC rises strongly, from roughly  $150\text{--}160 \text{ cm}^{-3}$  in the background case to about  $500 \text{ cm}^{-3}$ . The extra soluble particles act as additional cloud-condensation nuclei. They compete for the available water vapour, so the same condensate is distributed over many more droplets and the individual droplets are generally smaller.

**Exercise 8.** What is the effect of narrowing the sea-salt size distribution while keeping number concentration and median diameter fixed?

**Model answer.** The CDNC increases further, to around  $1100 \text{ cm}^{-3}$  in the supplied output. Narrowing the distribution around 100 nm removes much of the very-small-particle tail of the broad distribution. A larger fraction of the sea-salt population is therefore concentrated near sizes that activate readily at the supersaturations reached by the parcel.

**Exercise 9.** What is the effect of increasing the sea-salt median diameter from 100 nm to 800 nm with number concentration and width fixed?

**Model answer.** The particle-size spectrum shifts strongly towards much larger droplets and the CDNC diagnostic changes dramatically; in the supplied output the plotted CDNC falls to essentially zero while liquid water is still present. The coarse sea-salt particles activate and grow very efficiently and are rapidly shifted into much larger hydrometeor sizes, so they are no longer represented in the same cloud-droplet diagnostic category. The comparison therefore illustrates both the strong dependence of activation/growth on dry particle size and the importance of understanding exactly how a model diagnostic defines “cloud droplets”.



(a) Added 100 nm mode.

(b) Narrow distribution.

(c) 800 nm median diameter.

**Figure 4:** Sea-salt number, distribution width and median size all affect activation and the resulting droplet spectrum.

## 1.5 Bonus: ice formation

The 12 km experiment does not contain a formal exercise in the practical, but the intended interpretation is useful.

**Model answer.** Extending the parcel to 12 km allows it to cool to temperatures at which ice formation becomes important. The output shows a deep cloud, substantial changes in the particle-size distribution, and non-zero ice number at upper levels. The experiment demonstrates that a warm-cloud parcel model naturally becomes a mixed-phase/ice-cloud problem when the ascent is extended high enough.